

JunCheng Wu 吴俊成

Xi'an Jiaotong University | B.Eng. in Automation

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EDUCATION

Xi'an Jiaotong University

Xi'an, China

Bachelor of Engineering in Automation | Expected June 2028

- Rank: 11/188 (tied 10th; top 5.9%).
- Selected coursework: System Modeling and Dynamics Analysis (99), Numerical Computing Methods (96), Data Structures and Algorithms (95), Complex Functions and Integral Transforms (95), Programming Fundamentals (93), and Probability and Stochastic Processes (93).

RESEARCH EXPERIENCE

Refrigeration Compressor and Ultrasonic Sensing Research | Project Lead

2025

Provincial Undergraduate Innovation Project, Xi'an Jiaotong University

- Investigated non-invasive monitoring of transient refrigerant concentration in compressor oil using ultrasonic time-of-flight measurements.
- Performed theoretical validation of the apparatus and contributed to the construction and setup of the measurement instruments.
- Organized experimental results that supported a peer-reviewed conference paper at IIR TTPR 2025.

FlowFerry LLM Workflow Research | Research Assistant

Present

Intelligent Interaction Laboratory, Xi'an Jiaotong University | Advisor: Yunpeng Song

- Contributed to an execution-oriented framework for generating and evaluating platform-native LLM workflows.
- Trained experimental baselines and supported their evaluation for the AAAI 2027 submission.

PUBLICATIONS AND MANUSCRIPTS

FlowFerry: Generating and Evaluating Platform Native Workflows beyond Static Evaluation.

Yiheng Bian, Jiayou Zhang, **JunCheng Wu**, Lingyun Wang, Jiancheng He, Yunpeng Song, Zhongmin Cai.

Manuscript under review at AAAI 2027.

Monitoring the Transient Concentration of Refrigerant in Oil Using Ultrasonic TOF Method.

Che Wang, **JunCheng Wu**, Jianhua Wu. 7th IIR Conference on Thermophysical Properties and Transfer Processes of Refrigerants, 2025. [DOI](#)

SELECTED PROJECTS

Quadruped Robot Simulation Study

Ongoing

MuJoCo | Python/C++ | PyTorch | ROS 2

- Running and inspecting Unitree quadruped models in MuJoCo while studying simulation interfaces, model structure, and joint-space control.
- Working toward a reproducible simulation workflow while studying the progression from low-level control experiments to policy training.

TECHNICAL SKILLS

Programming Python, C/C++

Robotics / ML PyTorch, MuJoCo, ROS 2

Tools Linux, Git